KATARZYNA - S1222A EPB TBM 13,08m

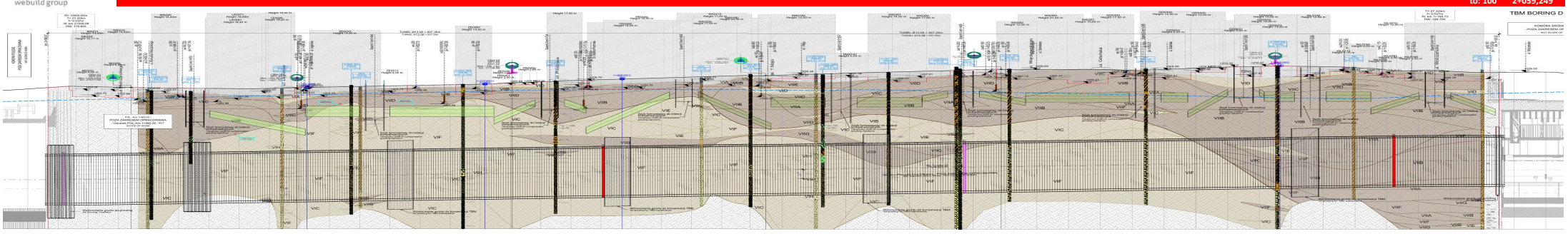
(POL-SRD) - K3 - Job Site: ODOLANOWSKA STATION LAUNCHING CHAMBER (Axix 23)

Improvement of the Łódź Railway Junction (TEN-T) Stage II, Łódź Fabryczna - Łódź Kaliska / Łódź Żabieniec section

TECHNICAL DATA ANALYSIS

Update: 27/07/2024

	Ring	Chainage
From:	1	1+954,734
to:	100	2+059,249



RING			POSITION OF THE CUTTERHEAD			EARTH PRESSURE BALANCE (EPB)			CH TORQUE			CH THRUST FORCE			GEOMETRICAL OF SOIL EXCAVATED			PRIMARY GROUT INJECTION		INJECTION OF FOAM						INJECTION OF BENTONITE			GEOMETRICAL OF MUCK EXTRACTED			PAT	
																				POLYMER		SURFACTANT		WATER	BENTONITE		WATER & BENTONITE	THEORETICAL OF CONDITIONED SOIL			TUNNEL AXIS 23 (POL-SRD)	
ID	Serial Number	Date of Instalation	CH initial [Km]	CH final [Km]	Length [m]	Average Value [bar]	Min. Value [bar]	Max. Value [bar]	Average Value [MNm]	Min. Value [MNm]	Max. Value [MNm]	Average Value [kN]	Min. Value [kN]	Max. Value [kN]	Average Density [Ton/m³]	Volume (V1) [m³]	Weight (W1) [Ton]	Expected Backfill [m³]	Injected Backfill [m³]	Volume (V2) [m³]	Weight (W2) [Ton]	Volume (V3) [m³]	Weight (W3) [Ton]	(V4) [m3] = (W4) [Ton]	Volume (V5) [m³]	Weight (W5) [Ton]	(V6) [m3] = (W6) [Ton]	(V) [m³] $\sum_{i=1}^{5}(V_i)$	(W) [Ton] $\sum_{i=1}^{6}(W_i)$	Density [Ton/m³]	Category of Soil	Description
															(LWK-000-W-MG-INP-09-1003_00)			1.233,72	1.418,47	1,1	[Ton/m³]	1,05	[Ton/m³]		2,55	[Ton/m³]						
1	D1331	21/08/2023	2+060,864	2+059,249	1,615	0,629	0,294	0,821	3,553	1,730	5,957	37.004	12.932	41.403	2,25	217,0	487,4	12,43	3,91	0,00	0,00	0,59	0,61	58,7	0,00	0,00	0,00	276,29	546,72	1,98	1% VIC 99% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
2	D0110	22/08/2023	2+059,248	2+057,696	1,552	0,711	0,408	0,840	3,939	2,223	6,868	37.402	9.072	40.952	2,24	208,6	467,6	11,94	4,82	0,00	0,00	0,56	0,59	56,1	0,00	0,00	0,00	265,23	524,31	1,98	2% VIC 98% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
3	D1782	30/08/2023	2+057,691	2+056,197	1,494	0,404	0,055	1,395	3,327	0,695	7,954	29.331	8.750	51.247	2,24	200,8	449,4	11,50	19,52	0,01	0,01	0,42	0,44	42,4	0,00	0,00	0,00	243,64	492,26	2,02	3% VIC 97% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
4	D1775	30/08/2023	2+056,194	2+054,586	1,608	1,499	1,068	1,935	7,801	2,801	11,158	48.955	18.208	51.665	2,23	216,1	481,4	12,37	13,33	0,00	0,00	0,40	0,42	40,2	0,00	0,00	0,00	256,67	522,05	2,03	5% VIC 91% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
5	D1772	30/08/2023	2+054,588	2+052,903	1,684	1,790	1,408	1,948	6,815	2,629	11,470	54.015	18.240	58.485	2,22	226,3	502,0	12,96	14,06	0,00	0,00	0,69	0,73	70,4	0,00	0,01	0,14	297,53	573,24	1,93	7% VIC 85% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
6	D1764	31/08/2023	2+052,903	2+051,380	1,523	1,589	0,743	1,864	9,536	3,198	13,009	54.335	18.144	57.166	2,21	204,7	452,0	11,72	17,03	0,00	0,00	0,44	0,46	44,1	0,01	0,02	0,27	249,50	496,83	1,99	9% VIC 79% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
7	D1693	31/08/2023	2+051,383	2+049,748	1,635	1,532	1,119	1,778	8,939	2,723	14,626	51.947	12.804	55.171	2,20	219,7	483,0	12,58	14,42	0,00	0,00	0,43	0,46	43,4	0,13	0,33	4,43	267,99	531,58	1,98	11% VIC 73% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
8	D1763	31/08/2023	2+049,748	2+048,004	1,744	1,547	1,185	1,796	8,041	2,787	13,262	48.995	21.007	55.815	2,19	234,4	512,9	13,42	13,19	0,00	0,00	0,41	0,43	41,3	0,22	0,55	7,33	283,40	562,43	1,98	13% VIC 67% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
9	D1688	31/08/2023	2+048,003	2+046,504	1,500	1,570	1,273	1,757	9,735	5,771	15,059	48.985	17.275	54.238	2,18	201,5	439,0	11,54	13,56	0,00	0,00	0,30	0,32	30,8	0,00	0,00	0,00	232,66	470,15	2,02	15% VIC 62% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
10	D1138	01/09/2023	2+046,508	2+044,877	1,631	1,568	1,228	1,851	10,546	4,305	14,103	46.836	21.071	51.729	2,17	219,1	475,1	12,55	13,46	0,00	0,00	0,31	0,32	30,9	0,00	0,00	0,00	250,32	506,36	2,02	17% VIC 56% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
11	D1784	01/09/2023	2+044,880	2+043,281	1,599	1,484	1,142	1,684	9,029	4,316	13,659	45.629	17.919	50.378	2,16	214,9	464,0	12,31	15,25	0,00	0,00	0,35	0,37	35,3	0,02	0,04	0,54	251,06	500,19	1,99	19% VIC 51% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
12	D1689	01/09/2023	2+043,283	2+041,729	1,553	1,560	1,055	1,958	9,104	1,918	14,962	49.314	19.495	54.914	2,15	208,7	448,7	11,95	14,38	0,00	0,00	0,30	0,31	30,2	0,13	0,32	4,31	243,52	483,81	1,99	21% VIC 46% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
13	D0085	02/09/2023	2+041,733	2+040,105	1,627	1,533	1,118	1,747	8,855	1,104	13,805	56.296	17.629	62.184	2,14	218,7	468,1	12,52	19,32	0,00	0,00	0,29	0,31	29,9	0,29	0,75	9,98	258,81	508,96	1,97	23% VIC 42% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
14	D1801	02/09/2023	2+040,102	2+038,482	1,620	1,564	1,340	1,910	9,949	2,335	16,019	56.488	21.071	60.769	2,14	217,7	465,8	12,47	12,90	0,00	0,00	0,27	0,29	27,3	0,01	0,03	0,39	245,73	493,87	2,01	22% VIC 1% VIF	Sands and silty sands. Sands and silty sands.
15	D0081	02/09/2023	2+038,482	2+036,865	1,616	1,537	1,230	1,796	10,826	4,755	15,031	56.782	22.004	60.769	2,13	217,2	463,5	12,44	12,79	0,00	0,00	0,29	0,30	29,0	0,03	0,08	1,01	247,51	493,90	2,00	22% VIC 2% VIF	Sands and silty sands. Sands and silty sands.
16	D1707	03/09/2023	2+036,860	2+035,267	1,593	1,547	1,252	1,831	11,261	4,371	16,158	56.364	15.409	60.769	2,13	214,1	455,8	12,26	14,96	0,00	0,00	0,26	0,27	26,4	0,12	0,31	4,15	244,84	486,94	1,99	22% VIC 3% VIF	Sands and silty sands. Sands and silty sands.
17	D1769	03/09/2023	2+035,267	2+033,714	1,553	1,564	1,302	1,766	9,054	3,315	13,159	52.736	18.176	57.648	2,13	208,7	443,4	11,95	13,08	0,00	0,00	0,30	0,31	29,6	0,00	0,00	0,00	238,54	473,30	1,98	22% VIC 4% VIF	Sands and silty sands. Sands and silty sands.
18	D1783	03/09/2023	2+033,715	2+032,115	1,600	1,574	1,379	1,729	10,486	4,750	14,115	51.193	19.238	55.236	2,12	215,0	456,6	12,31	14,08	0,00	0,00	0,28	0,30	28,2	0,00	0,00	0,00	243,51	485,12	1,99	21% VIC 5% VIF	Sands and silty sands. Sands and silty sands.
19	D1332	03/09/2023	2+032,118	2+030,513	1,605	1,573	1,385	1,732	9,693	4,119	12,290	48.378	17.372	52.340	2,12	215,7	457,0	12,35	13,54	0,00	0,00	0,30	0,31	30,1	0,00	0,00	0,00	246,17	487,48	1,98	21% VIC 6% VIF	Sands and silty sands. Sands and silty sands.
20	D1789	03/09/2023	2+030,510	2+028,876	1,633	1,621	1,410	1,806	10,208	3,658	13,175	48.106	18.884	53.724	2,11	219,5	463,7	12,57	13,40	0,00	0,00	0,28	0,29	28,3	0,02	0,05	0,69	248,72	493,05	1,98	21% VIC 7% VIF	Sands and silty sands. Sands and silty sands.
21	D1768	03/09/2023	2+028,870	2+027,328	1,543	1,589	1,360	1,821	10,007	3,502	13,042	50.912	18.208	54.560	2,11	207,3	437,0	11,87	12,97	0,00	0,00	0,27	0,28	27,0	0,04	0,10	1,34	235,95	465,75	1,97	21% VIC 8% VIF	Sands and silty sands. Sands and silty sands.
22	D1765	04/09/2023	2+027,330	2+025,683	1,647	1,618	1,290	1,780	10,476	3,350	14,878	53.885	17.211	58.807	2,11	221,3	466,4	12,67	15,00	0,00	0,00	0,31	0,32	31,4	0,01	0,03	0,40	253,38	498,57	1,97	20% VIC 9% VIF	Sands and silty sands. Sands and silty sands.
23	D1806	04/09/2023	2+025,684	2+024,130	1,554	1,670	1,478	1,844	10,069	4,435	12,971	50.539	15.860	55.364	2,10	208,8	439,4	11,96	13,44	0,00	0,00	0,29	0,31	29,6	0,00	0,00	0,00	238,66	469,30	1,97	20% VIC 10% VIF	Sands and silty sands. Sands and silty sands.
24	D0750	04/09/2023	2+024,131	2+022,557	1,574	1,657	1,386	1,918	10,542	2,786	15,586	51.697	18.080	58.742	2,10	211,6	444,5	12,11	12,90	0,00	0,00	0,27	0,29	27,6	0,02	0,05	0,71	240,12	473,12	1,97	20% VIC 11% VIF	Sands and silty sands. Sands and silty sands.
25	D1617	04/09/2023	2+022,560	2+020,999	1,562	1,655	1,494	1,932	10,821	2,996	15,255	57.987	23.323	64.533	2,10	209,8	440,0	12,01	12,86	0,68	0,75	0,25	0,27	25,7	0,05	0,13	1,72	238,15	468,53	1,97	20% VIC 12% VIF	Sands and silty sands. Sands and silty sands.
26	D1468	05/09/2023	2+021,002	2+019,362	1,640	1,635	1,462	1,922	12,072	3,011	16,536	60.440	20.428	67.396	2,10	220,3	462,8	12,61	14,33	0,13	0,14	0,27	0,29	27,6	0,02	0,04	0,55	248,84	491,36	1,97	19% VIC 12% VIF	Sands and silty sands. Sands and silty sands.
27	D1615	05/09/2023	2+019,366	2+017,750	1,616	1,657	1,241	1,863	9,514	2,724	15,858	62.703	21.811	70.420	2,10	217,2	456,2	12,44	13,48	0,00	0,00	0,34	0,35	35,1	0,22	0,57	7,58	260,16	499,76	1,92	19% VIC 12% VIF	Sands and silty sands. Sands and silty sands.
28	D1464	05/09/2023	2+017,742	2+016,171	1,572	1,694	1,489	1,824	12,641	4,862	21,667	61.514	20.878	64.629	2,10	211,2	443,2	12,09	13,10	0,05	0,05	0,29	0,31	29,6	0,00	0,00	0,00	241,06	473,15	1,96	19% VIC 12% VIF	Sands and silty sands. Sands and silty sands.
29	D1773	05/09/2023	2+016,163	2+014,547	1,615	1,638	1,472	1,898	11,767	2,640	17,462	63.656	20.524	68.908	2,10	217,1	456,1	12,43	13,27	0,00	0,00	0,38	0,40	39,2	0,10	0,26	3,43	260,13	499,41	1,92	18% VIC 12% VIF	Sands and silty sands. Sands and silty sands.
30	D1800	06/09/2023	2+014,547	2+012,953	1,594	1,609	1,365	1,867	9,799	1,817	20,712	72.001	23.227	86.537	2,10	214,1	448,6	12,26	20,06	0,06	0,07	0,45	0,47	42,5	0,64	1,62	21,66	278,78	514,89	1,85	18% VIC 14% VIF	Sands and silty sands. Sands and silty sands.
31	D1807	06/09/2023	2+012,954	2+011,327	1,627	1,679	1,407	1,847	10,094	3,014	18,500	62.483	19.463	67.203	2,09	218,6	456,7	12,52	13,78	0,00	0,00	0,45	0,47	30,1	0,20	0,51	6,80	255,97	494,57	1,93	18% VIC 16% VIF	Sands and silty sands. Sands and silty sands.
32	D1758	07/09/2023	2+011,323	2+009,733	1,590	1,663	1,473	1,894	13,999	2,575	22,443	61.112	18.498	67.042	2,08	213,7	444,9	12,24	14,45	0,01	0,01	0,44	0,47	29,3	0,10	0,26	3,50	247,00	478,52	1,94	20% VIC 16% VIF	Sands and silty sands. Sands and silty sands.
33	D1799	08/09/2023	2+009,732	2+008,107	1,625	1,586	0,379	1,868	8,795	2,411	15,869	68.410	21.972	86.248	2,08	218,3	452,9	12,50	18,34	0,16	0,17	0,61	0,65	60,6	0,92	2,35	31,30	310,99	548,05	1,76	22% VIC 16% VIF	Sands and silty sands. Sands and silty sands.
34	D1385	08/09/2023	2+008,107	2+006,538	1,569	1,687	1,502	1,813	13,426	2,943	19,747	77.286	20.524	88.113	2,07	210,9	436,1	12,07	13,37	0,04	0,05	0,44	0,46	28,3	0,46	1,16	15,49	255,16	481,54	1,89	24% VIC 16% VIF	Sands and silty sands. Sands and silty sands.
35	D1813	10/09/2023	2+006,540	2+004,926	1,615	1,699	1,322	1,965	11,436	1,811	16,714	70.622	22.004	81.132	2,06	216,9	447,1	12,42	15,38	0,12	0,13	0,58	0,61	36,5	0,59	1,50	20,01	274,17	505,89	1,85	26% VIC 16% VIF	Sands and silty sands. Sands and silty sands.
36	D1778	10/09/2023	2+004,930	2+003,338	1,592	1,589	1,292	1,863	6,936	2,036	15,603	78.797	25.832	93.003	2,10	213,9	448,7	12,25	13,40	0,22	0,24	0,63	0,66	58,3	0,68	1,74	23,22	296,27	532,88	1,80	28% VIC 16% VIF	Sands and silty sands. Sands and silty sands.
37	D1465	10/09/2023	2+003,338	2+001,718	1,619	1,645	1,082	1,893	10,911	1,497	19,649	69.221	24.739	78.012	2,05	217,6	445,3	12,46	13,65	0,46	0,51	0,49	0,52	49,6	0,55	1,39	18,57	286,69	515,85	1,80	31% VIC 15% VIF	Sands and silty sands. Sands and silty sands.
38	D1797	10/09/2023	2+001,715	2+000,132	1,583	1,648	0,945	2,136	12,714	5,559	20,528	65.132	18.916	76.564	2,04	212,7	433,8	12,18	14,92	0,45	0,49	0,50	0,52	49,8	1,22	3,11	41,51	304,91	529,19	1,74	33% VIC 15% VIF	Sands and silty sands. Sands and silty sands.
39	D1501	11/09/2023	2+000,132	1+998,524	1,608	1,656	0,852	2,171	13,754	3,407	21,033	62.668	21.071	70.516	2,03	216,0	439,0	12,37	13,86	0,54	0,59	0,33	0,34	33,0	1,07	2,74	36,48	286,33	512,09	1,79	36% VIC 14% VIF	Sands and silty sands. Sands and silty sands.
40	D1107	11/09/2023	1+998,523	1+996,912	1,611	1,660	0,964	1,956	11,073	3,305	18,868	67.225	20.299	75.728	2,03	216,5	438,3	12,39	13,32	0,42	0,46	0,51	0,53	60,2	0,90	2,29	30,56	308,16	532,40	1,73	38% VIC 14% VIF	Sands and silty sands. Sands and silty sands.
41	D1786	11/09/2023	1+996,915	1+995,318	1,597	1,680	1,294	1,946	11,705	4,444	18,672	64.872	20.331	72.350	2,02	214,6	433,0	12,29	13,27	0,59	0,65	0,28	0,30	45,8	0,69	1,77	23,61	284,91	505,14	1,77	41% VIC 13% VIF	Sands and silty sands. Sands and silty sands.
42	D1781	11/09/2023	1+995,315	1+993,732	1,583	1,595	1,144	1,872	12,403	2,258	19,337	70.489	25.639	78.495	2,01	212,7	427,6	12,18	14,49	1,04	1,15	0,28	0,29	35,2	1,44	3,68	49,08	298,30	517,02	1,73	43% VIC 13% VIF	Sands and silty sands. Sands and silty sands.
43	D1788	12/09/2023	1+993,739	1+992,111	1,628	1,672	1,322	1,984	10,480	3,747	22,532	71.290	20.428	76.018	2,00	218,7	438,1	12,52	13,73	0,70	0,77	0,47	0,50	59,4	2,72	6,93	92,45	371,77	598,18	1,61	46% VIC 12% VIF	Sands and silty sands. Sands and silty sands.
44	D1370	12/09/2023	1+992,113	1+990,532	1,581	1,645	1,090	1,944	13,078	2,401	19,934	69.352	20.042	79.717	2,00	212,5	425,6	12,17	13,35	0,24	0,26	0,45	0,47	46,2	1,44	3,68	49,12	308,53	525,40	1,70	46% VIC 12% VIF	Sands and silty sands. Sands and silty sands.
45	D1366	12/09/2023	1+990,532	1+988,913	1,619	1,664	0,735	2,007	13,102	4,041	22,619	66.366	20.396	72.157	2,00	217,5	434,8	12,45	14,93	0,54	0,59	0,55	0,58	55,3	1,84	4,70	62,63	336,56	558,63	1,66	48% VIC 11% VIF	Sands and silty sands. Sands and silty sands.
46	D1346	13/09/2023	1+988,916	1+987,306	1,610	1,660	1,243	2,085	9,951	1,482	22,802	75.395	23.098	85.154	2,00	216,4	432,5	12,39	13,73	0,76	0,83	0,59	0,62	59,5	2,69	6,87	91,60	368,77	591,88	1,61	48% VIC 11% VIF	Sands and silty sands. Sands and silty sands.
47	D1373	13/09/2023	1+987,308	1+985,690	1,619	1,542	0,958	1,892	13,665	3,159	22,401	70.321	16.053	77.176	2,00	217,5	434,0	12,46	13,77	0,74	0,81	0,17	0,18	17,4	0,64	1,62	21,60	257,44	475,59	1,85	50% VIC 10% VIF	Sands and silty sands. Sands and silty sands.
48	D1368	13/09/2023	1+985,696	1+984,132	1,564	1,557	0,978	1,983	12,087	0,961	23,266	75.114	22.583	89.046	2,00	210,2	420,0	12,03	13,46	0,45	0,49	0,29	0,30	29,2	3,15	8,04	107,23	347,38	565,33	1,63	49% VIC 10% VIF	Sands and silty sands. Sands and silty sands.
49	D1752	14/09/2023	1+984,134	1+982,482	1,651	1,625	1,332	1,843	13,296	4,429	20,933	72.224	21.972	79.331	2,00	221,9	444,2	12,71	13,90	0,12	0,13	0,22	0,23	22,2	0,84	2,13	28,47	272,85	497,35	1,82	49% VIC 9% VIF	Sands and silty sands. Sands and silty sands.
50	D1754	14/09/2023	1+982,488	1+980,897	1,591	1,583	0,921	1,901	11,800	0,991	23,034	77.224	14.959	83.384	2,01	213,7	428,7	12,24	13,59	0,30	0,33	0,18	0,19	18,7	0,84	2,14	28,49	261,37	478,47	1,83	48% VIC 9% VIF	Sands and silty sands. Sands and silty sands.
51	D1358	15/09/2023	1+980,907	1+979,213	1,694	1,552	0,564	1,925	10,911	1,075	22,490	77.446	20.042	84.768	2,01	227,6	457,3	13,03	18,83	0,37	0,40	0,51	0,54	53,8	1,99	5,09	67,80	350,12	584,95	1,67	48% VIC 8% VIF	Sands and silty sands. Sands and silty sands.
52	D1344	15/09/2023	1+979,215	1+977,731	1,484	1,571	1,177	1,929	12,339	1,322	22,059	67.565	16.728	78.623	2,01	199,4	401,6	11,42	13,62	0,07	0,08	0,71	0,74	74,9	0,42	1,08	14,36	289,49	492,83	1,70	47% VIC 8% VIF	Sands and silty sands. Sands and silty sands.
53	D1382	16/09/2023	1+977,738	1+976,131	1,607	1,489	1,277	2,018	12,733	1,012	20,663	67.361	19.752	75.278	2,02	215,9	435,8	12,36	12,63	0,04	0,05	0,96	1,01	96,4	0,08	0,21	2,78	316,09	536,27	1,70	47% VIC 7% VIF	Sands and silty sands. Sands and silty sands.
54	D1390	16/09/2023	1+976,124	1+974,547	1,577	1,204	0,901	1,454	10,603	2,135	21,980	57.902	16.825	68.425	2,02	211,9	428,8	12,13	13,19	0,00	0,00	0,71	0,75	71,8	0,23	0,58	7,73	292,21	509,73	1,74	46% VIC 7% VIF	Sands and silty sands. Sands and silty sands.
55	D1008	16/09/2023	1+974,546	1+972,925	1,621	1,218	0,942	1,400	8,747	2,041	14,411	52.585	21.522	59.932	2,03	217,8	441,8	12,47	13,41	0,00	0,00	0,60	0,63	61,2	0,15	0,38	5,11	284,69	509,03	1,79	46% VIC 6% VIF	Sands and silty sands. Sands and silty sands.
56	D1378	16/09/2023	1+972,925	1+971,297	1,628	1,155	0,917	1,257	6,263	2,098	12,594	51.246	17.082	57.037	2,03	218,7	444,6	12,52	13,00	0,00	0,00	0,45	0,47	44,8	0,00	0,00	0,00	263,95	489,90	1,86	45% VIC 6% VIF	Sands and silty sands. Sands and silty sands.
57	D1750	16/09/2023	1+971,296	1+969,742	1,554	0,973	0,651	1,191	9,150	1,905	16,473	51.612	20.557	57.005	2,04	208,8	425,3	11,95	12,68	0,00	0,00	0,28	0,29	28,3	0,00	0,00	0,00	237,38	453,98	1,91	45% VIC 5% VIF	Sands and silty sands. Sands and silty sands.
58	D1467	17/09/2023	1+969,733	1+968,111	1,622	0,938	0,693	1,067	8,986	3,675	16,320	42.700	16.439	53.724	2,04	217,9	445,0	12,48	14,41	0,00	0,00	0,17	0,18	20,2	0,04	0,09	1,26	239,52	466,70	1,95	44% VIC 5% VIF	Sands and silty sands. Sands and silty sands.
59	D1618	18/09/2023	1+968,105	1+966,558	1,547	0,899	0,679	1,255	11,325	1,425	17,319	50.528	17.404	62.506	2,05	207,9	425,4	11,91	15,18	0,00	0,00	0,26	0,28	33,9	0,29	0,73	9,72	251,77	470,00	1,87	44% VIC 4% VIF	Sands and silty sands. Sands and silty sands.
60	D1633	22/06/2024	1+966,475	1+964,345	2,129	0,524	0,055	1,367	4,860	0,789	13,805	42.566	0	154.416	2,05	286,1	586,7	16,38	23,59	0,00	0,00	0,77	0,81	71,2	5,52	14,07	187,55	545,65	860,33	1,58	43% VIC 4% VIF	Sands and silty sands. Sands and silty sands.
61	D1619	23/06/2024	1+964,350	1+962,740	1,609	1,108	0,056	1,735	5,854	0,772	13,987	58.171	15.409	70.838	2,05	216,2	444,3	12,38	20,41	0,00	0,00	1,12	1,18	101,3	0,00	0,00	0,00	318,63	546,71	1,72	43% VIC 3% VIF	Sands and silty sands. Sands and silty sands.
62	D1635	24/06/2024	1+962,741	1+961,150	1,591	1,626	1,233	1,867	8,899	1,349	16,330	69.109	25.253	74.602	2,06	213,8	440,2	12,24	15,52	0,01	0,01	1,49	1,56	111,5	0,00	0,01	0,10	326,90	553,37	1,69	42% VIC 3% VIF	Sands and silty sands. Sands and silty sands.
63	D1355	25/06/2024	1+961,148	1+959,512	1,636	1,599	1,179	1,960	10,543	1,302	21,423	70.754	22.551	81.518	2,06	219,8	453,4	12,59	16,95	0,28	0,31	1,33	1,40	79,7	0,39	0,99	13,15	314,22	548,94	1,75	42% VIC 2% VIF	Sands and silty sands. Sands and silty sands.
64	D1048	09/07/2024	1+959,511	1+957,946	1,565	1,682	1,070	2,199	10,867	0,640	22,692	65.178	322	77.530	2,07	210,3	434,9	12,04	12,46	0,20	0,22	2,75	2,89	119,9	22,81	58,18	775,68	1.108,83	1.391,71	1,26	41% VIC 2% VIF	Sands and silty sands. Sands and silty sands.
65	D1350	10/07/2024	1+957,946	1+956,326	1,619	1,777	0,549	2,025	3,763	0,466	15,618	75.224	16.407	80.843	2,07	217,6	450,8

KATARZYNA - S1222A EPB TBM 13,08m

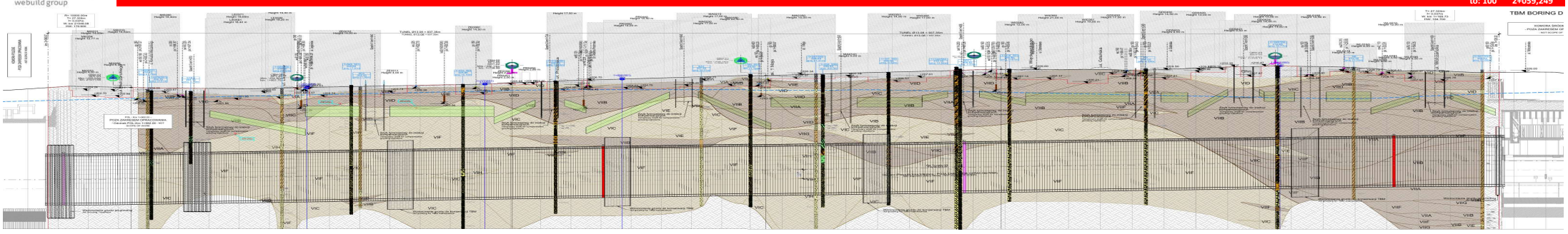
(POL-SRD) - K3 - Job Site: ODOLANOWSKA STATION LAUNCHING CHAMBER (Axix 23)

Improvement of the Łódź Railway Junction (TEN-T) Stage II, Łódź Fabryczna - Łódź Kaliska / Łódź Żabieniec section

TECHNICAL DATA ANALYSIS

Update: 27/07/2024

	Ring	Chainage
From:	1	1+954,734
to:	100	2+059,249



RING			POSITION OF THE CUTTERHEAD			EARTH PRESSURE BALANCE (EPB)			CH TORQUE			CH THRUST FORCE			GEOMETRICAL OF SOIL EXCAVATED			PRIMARY GROUT INJECTION		INJECTION OF FOAM					INJECTION OF BENTONITE			GEOMETRICAL OF MUCK EXTRACTED			PAT	
																				POLYMER		SURFACTANT		WATER	BENTONITE		WATER & BENTONITE	THEORETICAL OF CONDITIONED SOIL			TUNNEL AXIS 23 (POL-SRD)	
ID	Serial Number	Date of Instalation	CH initial [Km]	CH final [Km]	Length [m]	Average Value [bar]	Min. Value [bar]	Max. Value [bar]	Average Value [MNm]	Min. Value [MNm]	Max. Value [MNm]	Average Value [kN]	Min. Value [kN]	Max. Value [kN]	Average Density [Ton/m³]	Volume (V1) [m³]	Weight (W1) [Ton]	Expected Backfill [m³]	Injected Backfill [m³]	Volume (V2) [m³]	Weight (W2) [Ton]	Volume (V3) [m³]	Weight (W3) [Ton]	(V4) [m3] = (W4) [Ton]	Volume (V5) [m³]	Weight (W5) [Ton]	(V6) [m3] = (W6) [Ton]	(V) [m³] $\sum_{i=1}^{5}(V_i)$	(W) [Ton] $\sum_{i=1}^{6}(W_i)$	Density [Ton/m³]	Category of Soil	Description
															(LWK-000-W-MG-INP-09-1003_00)			1.233,72	1.418,47	1,1	[Ton/m³]	1,05	[Ton/m³]		2,55	[Ton/m³]						
80	D1320	22/07/2024	1+933,935	1+932,269	1,666	1,691	1,542	1,856	4,669	0,991	8,289	68.302	21.876	73.412	2,23	223,9	498,2	12,82	13,30	0,29	0,31	0,82	0,86	48,5	1,85	4,71	62,84	336,36	615,42	1,83	5% VIC 85% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
81	D1573	23/07/2024	1+932,261	1+930,715	1,547	1,739	1,627	1,897	4,322	0,841	7,131	66.462	17.436	69.133	2,23	207,8	462,4	11,90	13,44	0,26	0,29	0,70	0,73	23,4	0,80	2,05	27,36	259,60	516,32	1,99	5% VIC 85% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
82	D1585	23/07/2024	1+930,707	1+929,137	1,569	1,735	1,631	1,893	5,305	1,455	10,686	69.599	24.192	72.897	2,23	210,9	469,2	12,08	13,81	0,01	0,01	0,70	0,74	23,6	0,65	1,65	21,94	257,10	517,12	2,01	5% VIC 85% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
83	D0494	23/07/2024	1+929,119	1+927,563	1,556	1,838	1,708	1,962	7,113	1,492	11,374	72.326	22.101	75.438	2,23	209,1	465,3	11,98	14,96	0,18	0,20	0,52	0,55	17,5	1,08	2,75	36,73	264,02	523,01	1,98	5% VIC 85% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
84	D1347	23/07/2024	1+927,556	1+925,948	1,608	1,795	1,662	1,963	5,910	1,076	8,706	66.958	19.366	70.323	2,23	216,0	480,7	12,37	14,16	0,31	0,34	0,70	0,73	23,3	1,08	2,76	36,79	277,11	544,58	1,97	5% VIC 85% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
85	D1356	24/07/2024	1+925,943	1+924,303	1,640	1,828	1,614	1,965	6,248	0,769	9,123	65.165	16.696	69.873	2,23	220,4	490,3	12,62	17,43	0,28	0,31	0,75	0,79	25,1	0,98	2,50	33,37	279,89	552,39	1,97	5% VIC 85% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
86	D0493	24/07/2024	1+924,309	1+922,755	1,553	1,827	1,661	1,962	5,881	1,145	9,582	68.840	19.559	72.061	2,23	208,7	464,4	11,95	14,01	0,16	0,17	0,67	0,71	22,6	1,18	3,01	40,18	272,40	531,17	1,95	5% VIC 85% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
87	D1457	24/07/2024	1+922,758	1+921,144	1,614	1,890	1,745	1,993	6,702	1,029	10,431	68.420	18.819	72.704	2,23	216,8	482,7	12,42	14,00	0,28	0,30	0,60	0,63	22,4	1,20	3,06	40,84	280,94	549,91	1,96	3% VIC 2% VIF	Sands and silty sands. Sands and silty sands.
88	D1310	24/07/2024	1+921,148	1+919,543	1,605	1,864	1,706	2,008	5,778	1,205	9,133	72.426	20.878	75.181	2,23	215,7	480,4	12,35	18,72	0,07	0,08	0,67	0,70	35,3	0,99	2,52	40,31	292,03	559,29	1,92	1% VIC 4% VIF	Sands and silty sands. Sands and silty sands.
89	D1543	25/07/2024	1+919,542	1+917,952	1,591	1,856	1,584	2,001	4,109	0,580	9,662	78.103	20.074	82.966	2,23	213,7	476,1	12,24	14,74	0,34	0,37	0,61	0,64	31,9	1,02	2,61	41,70	288,27	553,29	1,92	5% VIF 85% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
90	D1045	25/07/2024	1+917,948	1+916,339	1,609	1,828	1,664	2,073	8,094	1,413	13,743	76.495	20.364	81.100	2,23	216,2	481,3	12,38	13,78	0,17	0,19	0,68	0,71	25,6	0,93	2,37	37,87	280,58	548,11	1,95	5% VIF 82% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
91	D0516	26/07/2024	1+916,333	1+914,754	1,578	1,750	1,500	2,017	2,743	0,489	9,234	86.451	28.502	90.784	2,22	212,1	471,8	12,14	18,24	0,32	0,35	1,23	1,29	63,5	0,90	2,30	40,81	317,93	580,00	1,82	5% VIF 79% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
92	D1565	26/07/2024	1+914,760	1+913,129	1,630	1,635	1,547	1,818	3,408	0,668	7,399	89.425	27.248	91.298	2,22	219,1	487,0	12,55	13,65	0,38	0,41	0,54	0,56	26,9	1,17	2,98	52,96	299,85	570,84	1,90	5% VIF 76% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
93	D1050	26/07/2024	1+913,130	1+911,534	1,595	1,628	1,473	1,998	6,322	1,081	8,324	87.608	21.071	89.658	2,22	214,4	476,2	12,27	12,66	0,00	0,00	0,44	0,46	22,1	0,82	2,09	37,12	273,98	537,92	1,96	5% VIF 73% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
94	D0399	26/07/2024	1+911,529	1+909,937	1,591	1,646	1,481	1,983	5,082	0,932	9,015	84.001	22.294	87.438	2,22	213,8	474,7	12,24	13,10	0,00	0,00	0,49	0,52	24,8	1,04	2,65	47,10	286,27	549,81	1,92	5% VIF 70% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
95	D1321	26/07/2024	1+909,938	1+908,326	1,612	1,673	1,265	1,884	5,805	1,470	9,909	77.783	27.377	80.457	2,22	216,6	480,5	12,40	14,36	0,00	0,00	0,44	0,46	22,0	0,69	1,77	31,41	270,42	536,12	1,98	5% VIF 67% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
96	D1770	27/07/2024	1+908,324	1+906,730	1,594	1,683	1,551	1,908	7,138	1,324	10,172	75.566	22.197	79.620	2,22	214,2	475,0	12,27	12,98	0,11	0,12	0,38	0,40	19,1	0,73	1,87	33,17	267,04	529,66	1,98	5% VIF 64% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
97	D1674	27/07/2024	1+906,712	1+905,126	1,586	1,677	1,559	1,972	6,863	1,253	10,456	74.510	23.323	78.623	2,22	213,1	472,1	12,20	15,05	0,16	0,17	0,57	0,60	29,2	1,29	3,28	58,35	301,35	563,67	1,87	5% VIF 61% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
98	D1539	27/07/2024	1+905,129	1+903,527	1,603	1,643	1,545	1,939	6,631	2,101	9,513	76.818	23.838	80.972	2,21	215,4	476,8	12,33	16,69	0,18	0,20	0,50	0,52	25,2	0,75	1,92	34,05	275,31	538,71	1,96	5% VIF 58% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
99	D1555	27/07/2024	1+903,519	1+901,918	1,601	1,648	1,585	1,930	4,808	1,381	6,615	75.492	21.682	78.848	2,21	215,2	476,1	12,32	14,58	0,00	0,00	0,50	0,52	25,1	1,12	2,87	50,95	291,70	555,51	1,90	5% VIF 55% VIIA	Sands and silty sands. Clay and sandy loam, rigid.
100	D1677	27/07/2024	1+901,917	1+900,318	1,599	1,668	1,570	1,901	3,199	0,624	5,259	74.942	20.267	78.237	2,21	214,8	474,9	12,30	14,12	0,09	0,10	0,57	0,60	28,4	0,92	2,35	41,78	285,65	548,15	1,92	5% VIF 52% VIIA	Sands and silty sands. Clay and sandy loam, rigid.